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Descriptor: *The SFI Smart Ocean Dataset for Acoustic Communications (SODAC)*

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ABSTRACT Underwater acoustic data communication is an enabling technology for operations such as environmental monitoring, exploitation of ocean resources, and autonomous inspection of offshore infrastructure. The only viable technique for subsea telemetry over longer distances is by encoding the information in sound waves. However, the influence of the ocean environment on sound propagation can be harsh, resulting in a poor performance of protocols designed for terrestrial radio-frequency communications. This justifies active research on physical-layer algorithms and network protocols that are robust to acoustic propagation in the oceans. Factors that hamper progress are the lack of accepted standard test channels and hard-to-obtain field data. A majority of academic institutions cannot afford at-sea experimentation, resorting to simplified channel models to test new protocols. To make field data accessible to a wider audience, this article presents a dataset of hydrophone recordings of communication and channel probe waveforms collected in an enclosed fjord environment. The SFI Smart Ocean Dataset for Acoustic Communications (SODAC) allows users to test communication algorithms on *in situ* data, with opportunities to publish reproducible results that set a realistic benchmark for other researchers.

IEEE SOCIETY/COUNCIL Oceanic Engineering (OES)

DATA TYPE/LOCATION Hydrophone time series; Austevoll, Norway

DATA DOI/PID 10.21227/3aa6-4k33

INDEX TERMS Acoustic communication, hydrophone data, signal processing.

BACKGROUND

In situ testing of underwater acoustic communication solutions requires resources unavailable to the wider research community. Existing datasets are usually shared with a small number of collaborators and are shelved after completion of the funding project. Furthermore, large differences between ocean environments preclude meaningful comparisons of at-sea system performances reported by different authors. These challenges have stimulated the development of channel models based on acoustic propagation modeling and statistical approaches [1], [2], [3], [4]. Alternatively, channel replay simulation has emerged as a benchmark for testing modulation schemes in measured channels [5]. All these channel models are important steps forward, but none of them can capture all propagation and scattering effects that can occur in acoustic channels.

SODAC is a dataset collected within SFI Smart Ocean, a Centre for Research-Based Innovation which is developing a smart and wireless underwater sensor network, for the benefit of science and industry [6]. The bulk of the dataset consists of acoustic time series recordings, collected at a marine aquaculture research facility in the southwest of Norway [7]. Data were collected in three frequency regimes encompassing bands A, B, and E of the JANUS standard [8] and feature an assortment of communication and channel probing waveforms.

The reuse potential of the data includes the following.

1) Test and compare communication receiver algorithms on *in situ* signal recordings. This potential is leveraged by the presence of waveforms with a flexible interpretation in terms of spectral efficiency and forward error correction coding [9].
2) Evaluate channel estimation methods on *in situ* probe signal recordings.
3) Extract replay channels for channel simulators.
4) Extract error statistics for network simulations.
5) Study ambient noise characteristics.
6) Ground truth for acoustic modeling.

At the time of writing, the only existing publication made with SODAC is the green replay channel of [10].

COLLECTION METHODS AND DESIGN

The data were collected at the Austevoll marine aquaculture research station of the Institute of Marine Research in Austevoll municipality, southwest of Bergen, Norway [7]. The center of operations was the floating cage aquaculture platform of the station (Fig. 1). For further descriptions of the surroundings and bathymetry, see [7].

Acoustic Data

Transmitter (Tx) hardware consisted of a Neptune Sonar T170 transducer, a Teledyne Benthos ATM-900 series acoustic modem, and a NORCE software-defined modem equipped with a Kongsberg TD30H transducer. These senders operate in different frequency bands, designated low frequency (LF), mid frequency (MF), and high frequency (HF) in Table I. The table has two entries for the source level. The first one is the expected source level based on product documentation and instrument settings. The second entry is an estimated value obtained with a reference hydrophone deployed from the aquaculture research platform.

Fig. 1 sketches the experiment geometry. The transmitters were suspended in the water column from the deep end of the platform to a depth of ≈20 m, which is the maximum cable length of the MF modem. They were deployed close together with a horizontal separation of a few meters. Two receive hydrophones, R1 and R2, were mounted on bottom tripods. Their locations were selected to ensure a line of sight to the senders and to yield different ranges. R1 was placed at close range, on the top of a subsea hill, and R2 further away on a hillside. Instrument positions and transmission ranges are given in Table II.

TABLE I. Sender Details

Band	LF	MF	HF
Hardware	T170	ATM-900	TD30H
Depth (m)	20	19	20
Center freq. (Hz)	6000	11 520	28 000
Bandwidth (kHz)	4.5	5.6	9.0
Expected source level (dB re 1 μPa^{2}m^2)	181	176	163
Measured source level (dB re 1 μPa^{2}m^2)	183	168	160

Note: The bandwidth is the −3 dB value of the probing waveforms.

TABLE II. Instrument Locations and Depths, and Transmission Ranges

	Latitude (°N)	Longitude (°E)	Depth (m)	Range (km)
Tx	60.08888	5.26785	19–20	—
R1	60.09109	5.26571	43	0.27
R2	60.08478	5.28424	14	1.02

SODAC spans the period from 14-Nov-2024 17:00 to 15-Nov-2024 06:20 (UTC), where each of the senders transmitted a waveform sequence once every 10 min. The dataset contains 80 recorded waveform cycles for each band, containing channel probe and communication waveforms of the following types.

1) Linear frequency modulated (LFM) chirp train.
2) Aperiodic binary-phase shift keyed (BPSK) pseudorandom binary sequence (A-PRBS).
3) Periodic BPSK-modulated pseudorandom binary sequence (P-PRBS).
4) Bandpass filtered white Gaussian noise (BFGN).
5) Multitone waveform.
6) JANUS (binary frequency-shift keying, compliant with the standard [8]).
7) Orthogonal frequency-division multiplexing (OFDM).
8) Orthogonal signal-division multiplexing (OSDM). This modulation technique is also known as orthogonal time frequency space (OTFS) modulation in radiofrequency literature [12].
9) Multiple frequency-shift keying (MFSK).
10) Single-carrier quadrature phase-shift keying (SC-QPSK).
11) Sweep-spread carrier quadrature phase-shift keying (S2C-QPSK).
12) Binary chirp slope keying (BCSK).
13) Frequency-hop binary chirp slope keying (FH-BCSK).

Some waveform types are present in all bands, and some only in one or two bands. The variety is largest for LF, whose waveform sequence was altered between cycles 13 and 14. Table III summarizes the presence of waveforms in the experiment data.

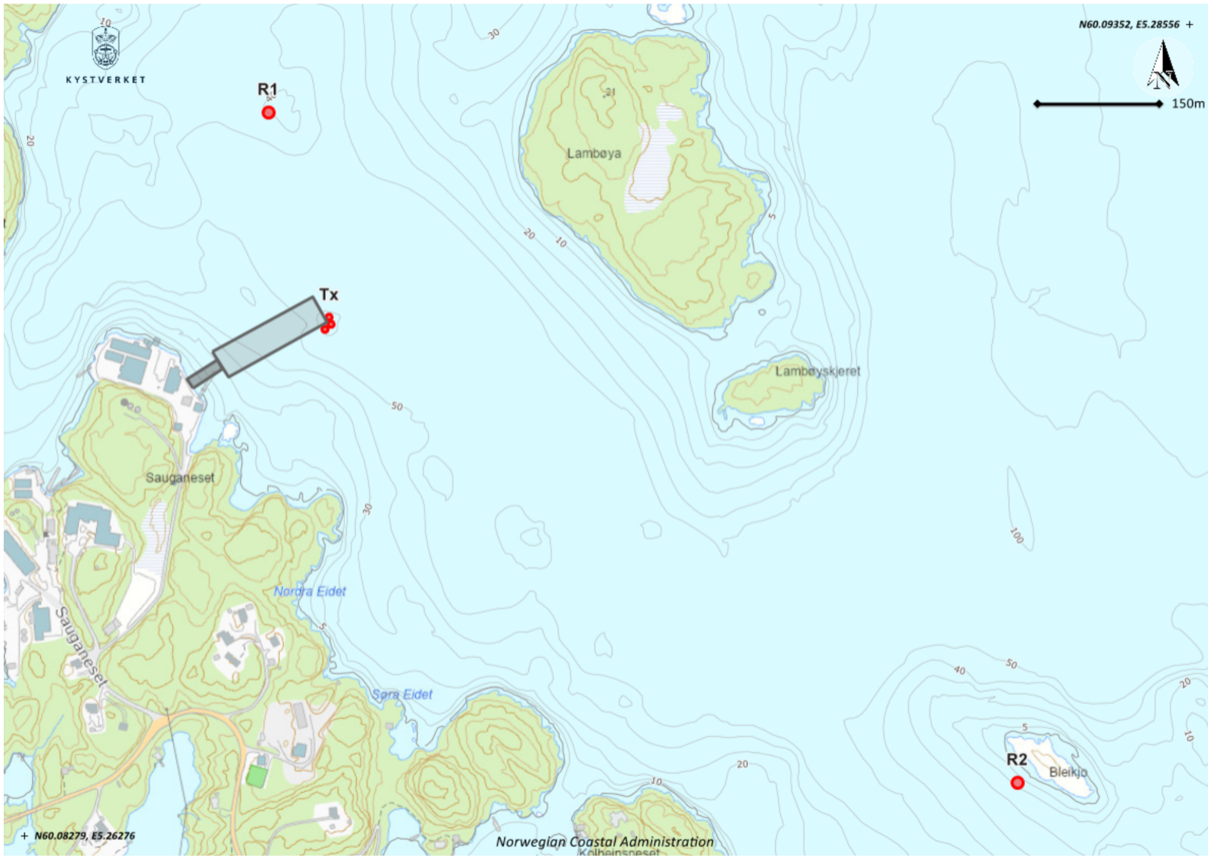

FIG. 1. **Area map generated with the help of [11]. The three transmitters are deployed from the deep end of the aquaculture platform (gray-shaded); the two receivers (R1 and R2) stand on the seafloor.**

TABLE III. **Presence of Waveform Types in the Acoustic Data**

	LF 1–13	LF 14–80	MF	HF
LFM	X	X	X	X
A-PRBS	X	X	X	X
OFDM	X	X	X	X
OSDM	X	X	X	X
MFSK	X	X	X	X
P-PRBS	X			
BFGN	X			
Multitone	X			
JANUS		X	X	X
SC-QPSK			X	X
S2C-QPSK	X	X		
BCSK	X	X		
FH-BCSK	X	X		

Note: The LF signal sequence was changed between cycles 13 and 14.

As a security precaution, the hydrophone recordings were filtered to replace ambient noise below ≈2.5 kHz with synthetic noise. Details of this filter are not provided, since it does not affect any of the signaling bands.

Nonacoustic Data

SODAC is primarily a dataset for communications and signal processing, but some environment information was collected in support of acoustic modeling. A SAIV A/S CTD (conductivity, temperature, depth) profiler Model SD204 was used to measure sound speed profiles. This was done twice, near the start and the end of the acoustic experiment, from the deep end of the platform close to the transmitters.

Weather conditions were calm, with little wind and some drizzle. A photograph of the sea surface is available, taken at the end of the experiment, pointing from the platform in the direction of R2.

SODAC includes bathymetry data of Vestland County downloaded from Geonorge [13]. The same bathymetry data were used to generate the 3-D model shown in Fig. 4 of [7].

VALIDATION AND QUALITY

The soundscape at Austevoll differs from an open ocean environment. Ambient noise at the R1 location is influenced by the proximity of the research station, where machineries and monitoring instruments emit sound at specific frequencies. Noise received on R2 contains wideband impulsive clicks, presumably self-noise of the receiver platform. Fig. 2

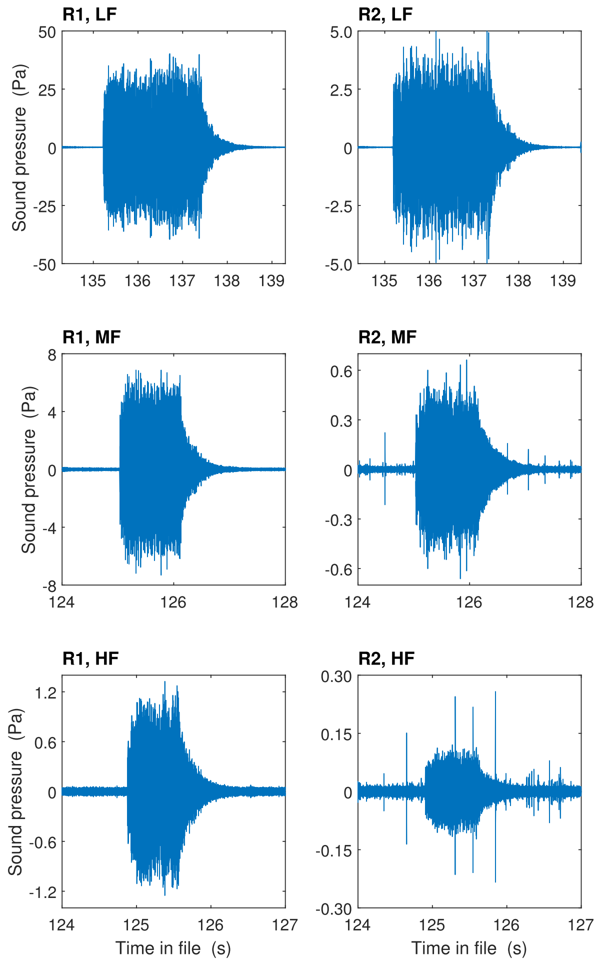

FIG. 2. **In-band time series of JANUS packets received in experiment cycle 50.**

illustrates the quality of the acoustic data. These time series are bandpass filtered to remove out-of-band noise and reveal a high overall signal-to-noise ratio (SNR), except in the HF band on R2, where clicks interfere with the reception. Clicks are also noticeable in the MF band on R2, but here they are weaker relative to the signal strength. The high overall SNR is characteristic of the entire experiment, whereas the severity of the clicks varies between cycles.

LF and MF recordings reveal some out-of-band signal power due to spectral sidelobes and nonlinear hardware responses, but the senders transmitted their waveform sequences in different time slots. Hence, out-of-band power does not interfere with the signal reception in other bands.

Table IV gives an overall assessment of the data quality, where the used criterion is whether the recordings are a faithful representation of the underwater soundscape. The only significant cause of degradation is the impulsive noise on R2, which does not belong to the environment. These clicks are unimportant for LF due to the high overall SNR, may affect a few MF signal receptions, and can be destructive for the HF band.

TABLE IV. **Quality of the Acoustic Data**

	R1	R2
LF	High	High
MF	High	Good
HF	High	Questionable

Note that the data quality is judged purely from an experimental technical perspective. A high quality does not imply benign conditions for underwater acoustic communications. As a case in point, the gradually decaying tail of the packets in Fig. 2 is a sign of reverberation, which is a form of multipath propagation due to a myriad of scatterers. Communication systems are limited by reverberation in this dataset, not by noise (with the possible exception of HF on R2).

RECORDS AND STORAGE

The hydrophone data are stored as 24-bit wave files with self-explanatory file names
LF_R1_cycle01.wav
LF_R1_cycle02.wav
...
LF_R1_cycle80.wav
and similar for MF, HF, and R2. Each wave file header contains a timestamp that gives the start of the recording. The sampling rate is 96 kHz and the time interval between successive cycles is 10 min. The duration of the wavefiles is 193 s for LF cycles 1–13, 151 s for LF cycles 14–80, 131 s for MF, and 130.4 s for HF. Fig. 3 shows example spectrograms for the first cycle of each type received on R1. The figure includes the name of the wave file, the timestamp, and the signal acronyms of Table III. The name LFM-225 denotes an LFM train with a pulse repetition time of 225 ms, and analogously for the other LFM probes [14]

The dataset includes wave files of the transmitted signals, with file names that are in accordance with Table III and Fig. 3. Table V lists all transmit signals, a technical description of which is provided as supplementary information, published alongside this article on IEEE Xplore [14]. Different sampling rates occur in Table V, as some waveforms are easier to generate at a specific rate; e.g., an integer number of samples per symbol may be required. To facilitate the validation of transmitter implementations, the waveforms are provided at a "native" sampling rate.

The file Basisdata_46_Vestland_25832_Dybdedata_FGDB.zip holds a file geodatabase (FGDB) with feature layers of the bathymetry.

Finally, sound speed profiles are provided in the file CTD.csv with columns representing date, time (UTC), salinity (ppt), temperature (°C), depth (m), and sound speed (ms^{-1}). The photograph of the sea surface is named seasurface.jpg.

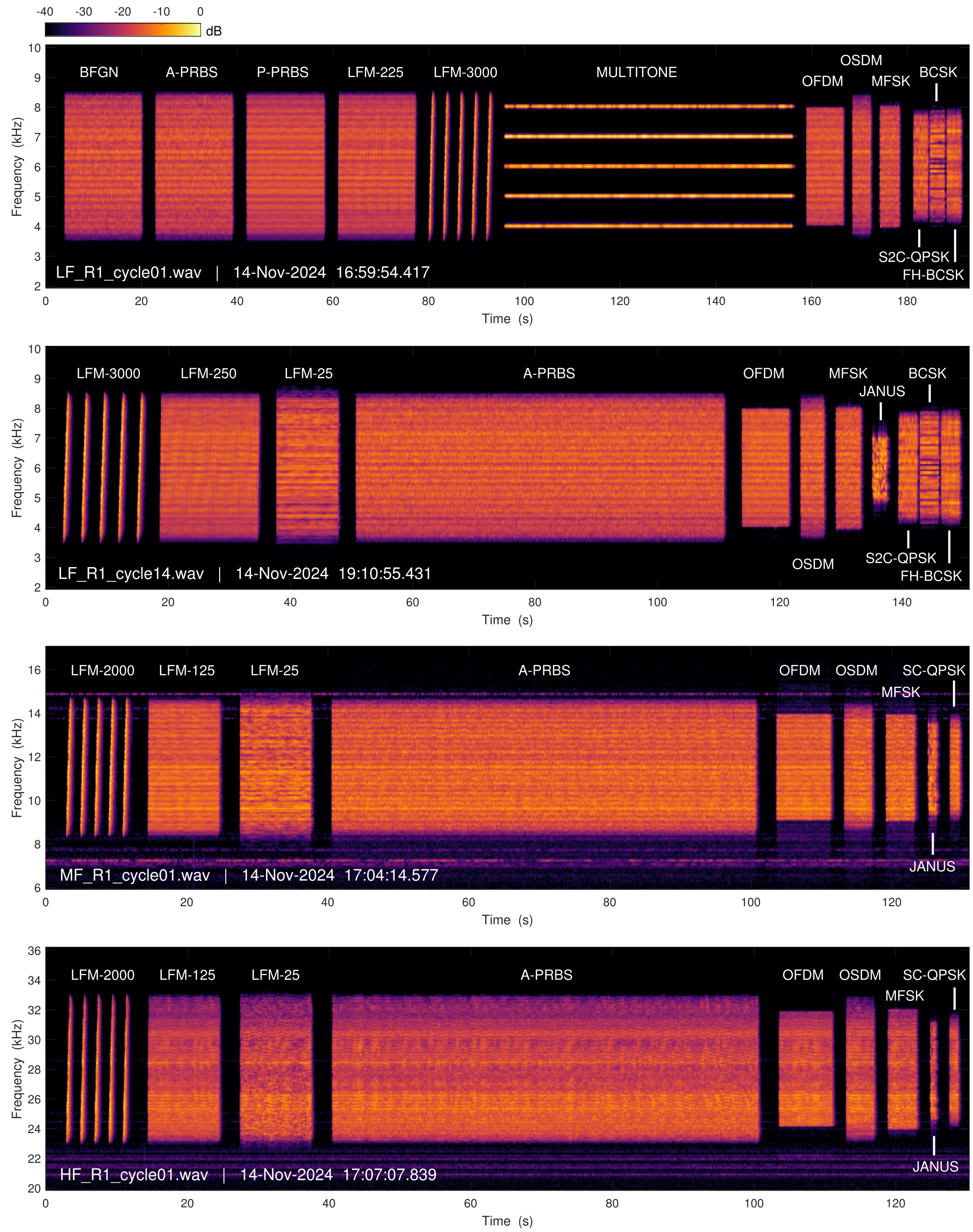

FIG. 3. Cycle spectrograms with signal types. Counted from the top, the first graph is illustrative of LF cycles 1–13, the second graph of LF cycles 14–80, the third graph of MF cycles 1–80, and the fourth graph of HF cycles 1–80.

INSIGHTS AND NOTES

Users of the dataset may find the following information and reminders useful.

1) The senders were deployed from the far end of the research platform, at a depth below the aquaculture fish cages. There is a free line of sight to the receivers.
2) The A-PRBS waveform can be used for channel estimation using the known transmitted bit sequence, but it can also be interpreted as a communication waveform with flexible coding schemes and symbol constellations [9]. Several of the other waveforms (e.g., OFDM, OSDM, and MFSK) also have vanilla designs which permit a flexible interpretation of the spectral efficiency.
3) The waveform sequences transmitted at sea contain proprietary signals that are not included in SODAC. The owners of these waveforms can compare their proprietary results with those published for the open dataset.
4) The signals have different root-mean-square values in the wave files of Table V, but they were transmitted

TABLE V. **Transmit Waveforms Details**

File Name	Length (s)	Sampling Rate (Hz)
LFM-25-LF.wav	10.0	90000
LFM-225-LF.wav	16.0	90000
LFM-250-LF.wav	16.0	90000
LFM-3000-LF.wav	15.0	90000
A-PRBS-LF-16s.wav	16.0	90000
A-PRBS-LF.wav	60.0	90000
P-PRBS-LF.wav	16.2	90000
MULTITONE-LF.wav	60.0	96000
BFGN-LF.wav	16.0	96000
OFDM-LF.wav	7.6	96000
OSDM-LF.wav	3.7	96000
MFSK-LF.wav	4.0	96000
JANUS-LF.wav	2.2	96000
S2C-QPSK-LF.wav	3.0	96000
BCSK-LF.wav	3.0	96000
FH-BCSK-LF.wav	3.0	96000
LFM-25-MF.wav	10.0	90000
LFM-125-MF.wav	10.0	90000
LFM-2000-MF.wav	10.0	90000
A-PRBS-MF.wav	60.0	90000
OFDM-MF.wav	7.6	96000
OSDM-MF.wav	3.8	96000
MFSK-MF.wav	4.0	96000
JANUS-MF.wav	1.1	96000
SC-QPSK-MF.wav	1.3	96000
LFM-25-HF.wav	10.0	90000
LFM-125-HF.wav	10.0	90000
LFM-2000-HF.wav	10.0	90000
A-PRBS-HF.wav	60.0	90000
OFDM-HF.wav	7.5	96000
OSDM-HF.wav	3.9	96000
MFSK-HF.wav	4.0	96000
JANUS-HF.wav	0.7	96000
SC-QPSK-HF.wav	1.2	96000

at the same source level (differing per band, see Table I).

5) Noise below ≈2.5 kHz has been replaced by somewhat realistic looking synthetic noise.
6) The header of the wave files contains a timestamp for the start of the recording.
7) The MF sender has a noticeable clock frequency offset that causes an apparent Doppler shift corresponding to a Doppler velocity of about 0.4 ms^{-1} (increasing range).
8) The receivers have been calibrated for frequencies up to 20 kHz. Data can be converted to pascals with a multiplication factor of 94. (This assumes a wave file conversion with clipping values of -1 and 1.) The conversion in Fig. 2 uses this factor also for the HF band despite the uncertain calibration above 20 kHz.

SOURCE CODE AND SCRIPTS

The JANUS signals were generated using the open-source toolkit available from [15]. There are no public repositories containing source code for the other signal types, but these are specified in [14].

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All authors reviewed the manuscript. The authors have declared no conflicts of interest.

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